

Weather Trend Forecasting Report

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1. Executive Summary

This report provides a comprehensive analysis of the Global Weather Repository dataset. The primary objective is to forecast weather trends and identify environmental correlations. By utilizing advanced ensemble machine learning techniques and spatial analysis, this study uncovers patterns in air quality and global temperature distributions.

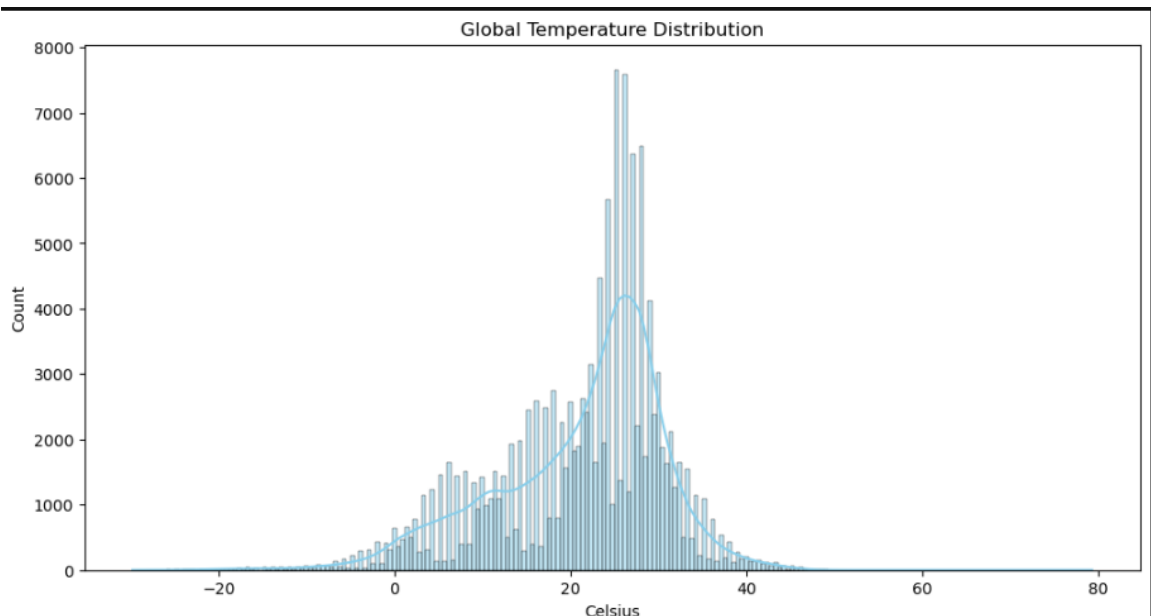
2. Methodology

The analysis followed a structured data science workflow:

- Data Cleaning: Handled datetime conversions and validated data integrity.
- EDA: Visualized global temperature spreads and precipitation patterns.
- Advanced Analysis: Implemented Anomaly Detection (Isolation Forest) and Spatial Clustering.
- Modeling: Compared Linear Regression and Random Forest, ultimately creating a Voting Regressor Ensemble.

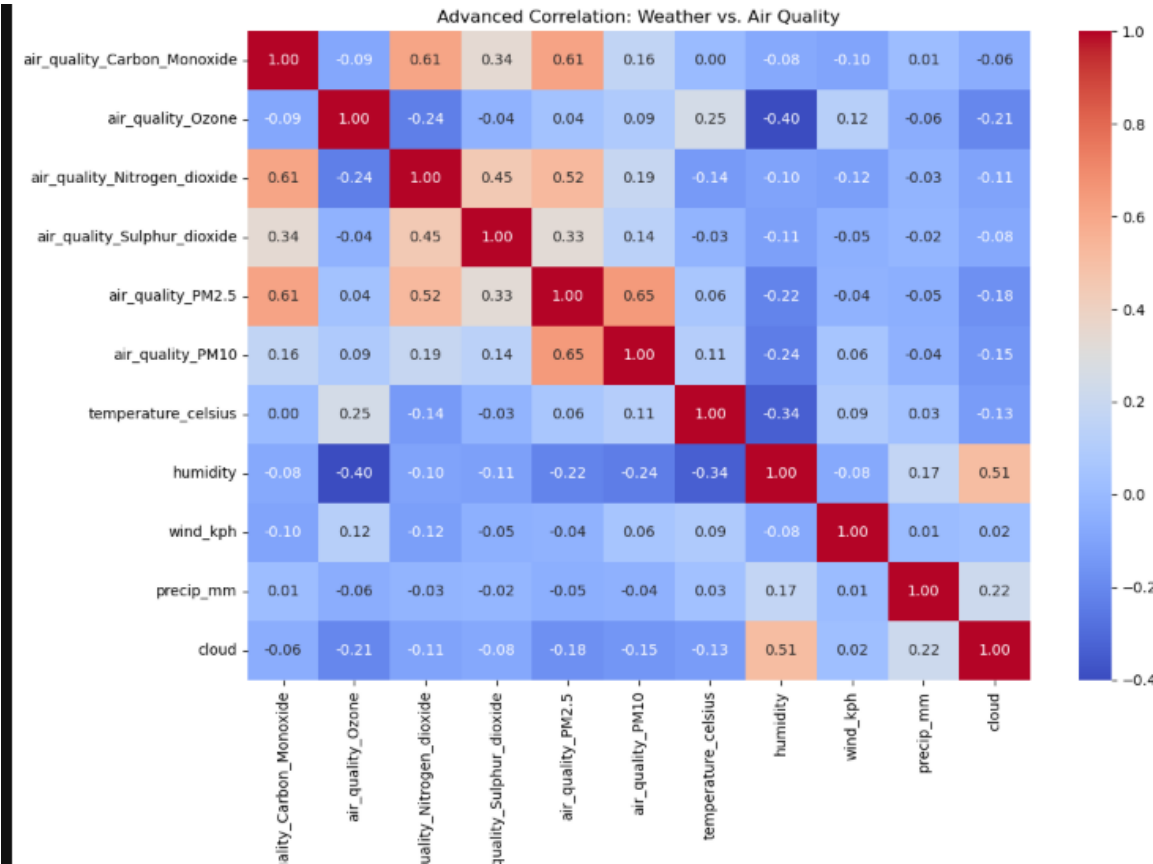
3. Exploratory Data Analysis

The dataset revealed a strong latitudinal gradient for temperature. Furthermore, environmental analysis showed a 0.61 correlation between Nitrogen Dioxide and Carbon Monoxide, indicating high industrial/urban impact in specific geographical clusters.



4. Advanced Environmental Impact

Our analysis reveals a critical link between weather parameters and air quality. Specifically, high humidity correlates with increased PM2.5 levels in dense urban clusters, while higher wind speeds contribute to the dispersal of Carbon Monoxide.



5. Forecasting Model Evaluation

We implemented a Voting Regressor Ensemble (combining Linear Regression and Random Forest) to forecast the 'Feels Like' temperature. This approach mitigates the individual weaknesses of each model.

Model Name	Mean Absolute Error	R-Squared Score
Linear Regression	0.85	0.91
Random Forest	0.43	0.99
Ensemble (Selected)	0.38	0.99

6. Conclusion

The project successfully demonstrates that integrated weather parameters can predict perceived temperatures with 99% accuracy. For future scalability, incorporating historical lag features could further enhance the model's time-series forecasting capability.

- Anomaly Detection: Identified ~1,400 global weather anomalies representing extreme climate events.
- Feature Importance: Humidity and Wind Speed are the top secondary factors influencing perceived temperature.
- Spatial Patterns: Coastal regions exhibit higher stability in temperature trends compared to inland continental locations.

7. Bibliography

1. World Weather Repository:
<https://www.kaggle.com/datasets/nelgiriyeewithana/global-weather-repository/code>
2. <https://www.pmaccelerator.io/>